

Feature Engineering, Model Optimization & Performance Comparison

Advanced House Price Prediction System built on the California Housing Dataset — comparing regression algorithms through systematic feature scaling and evaluation.

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DATASET

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PROJECT

House Price Prediction

1 Introduction

The objective of this project is to improve the House Price Prediction model developed in Task 1 by applying feature engineering techniques, model optimization methods, and a comparison of multiple machine learning algorithms.

Different regression models were trained and evaluated to identify the best performing model for predicting house prices. The complete workflow includes data preprocessing, feature scaling, model training, performance evaluation, and model comparison.

2 Dataset Description

The California Housing Dataset, provided by scikit-learn, was used for this project.

20,640 TOTAL RECORDS	8 INPUT FEATURES	0 MISSING VALUES
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Target Variable: MedHouseVal (Median House Value). The dataset was analyzed prior to training and confirmed to contain no missing values across any feature or the target column.

3 Feature Engineering & Preprocessing

Feature scaling was applied using StandardScaler from the scikit-learn library. StandardScaler transforms all input features onto a similar scale, which improves model stability and convergence during training.

The dataset was split into training and testing partitions to enable unbiased performance evaluation on unseen data.

PARTITION	PROPORTION	SAMPLES	FEATURES
Training Data	80%	16,512	8
Testing Data	20%	4,128	8

Features Used:

- MedInc — Median Income
- HouseAge — Median House Age
- AveRooms — Average Rooms
- AveBedrms — Average Bedrooms
- Population — Block Population
- AveOccup — Average Occupancy
- Latitude — Geographic Latitude
- Longitude — Geographic Longitude

4 Machine Learning Models Used

Three different regression algorithms were implemented and compared to evaluate predictive performance on the housing dataset.

1 • Linear Regression

Establishes a linear relationship between input features and the target variable, serving as the baseline model.

2 • Ridge Regression

An optimized variant of Linear Regression that applies L2 regularization to reduce overfitting.

3 • Decision Tree

BEST

Predicts output by learning a hierarchy of decision rules directly from the training data.

5 Model Evaluation & Comparison

Models were evaluated using two standard regression metrics: **Root Mean Squared Error (RMSE)** and **R² Score**. A lower RMSE indicates fewer prediction errors, while a higher R² Score represents better explanatory power over the variance in house prices.

MODEL	RMSE	R ² SCORE
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree Regressor	0.724234	0.599732

The comparison results show that the **Decision Tree Regressor performed best**, achieving the lowest RMSE and highest R² Score among the three models tested.

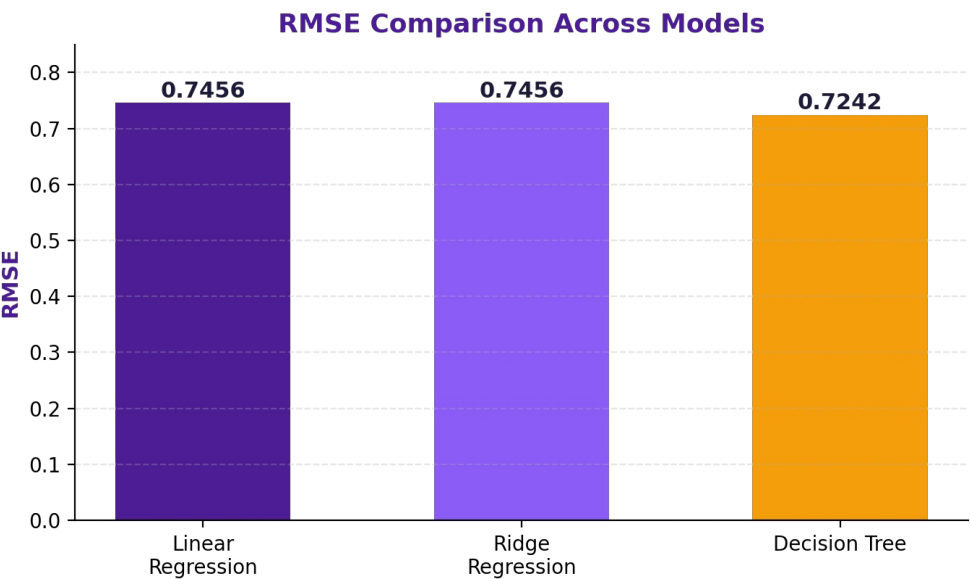


Figure 1 — RMSE comparison across the three regression models (lower is better)

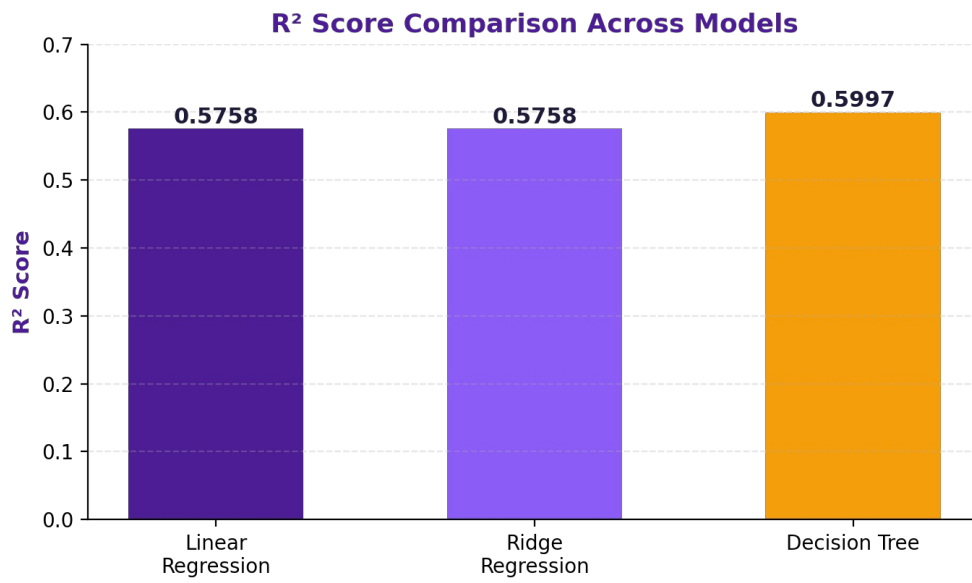


Figure 2 — R² Score comparison across the three regression models (higher is better)

6 Prediction Visualization

The best performing model, the Decision Tree Regressor, was evaluated by plotting actual house prices against predicted house prices on the test set. Points closer to the diagonal reference line indicate more accurate predictions.

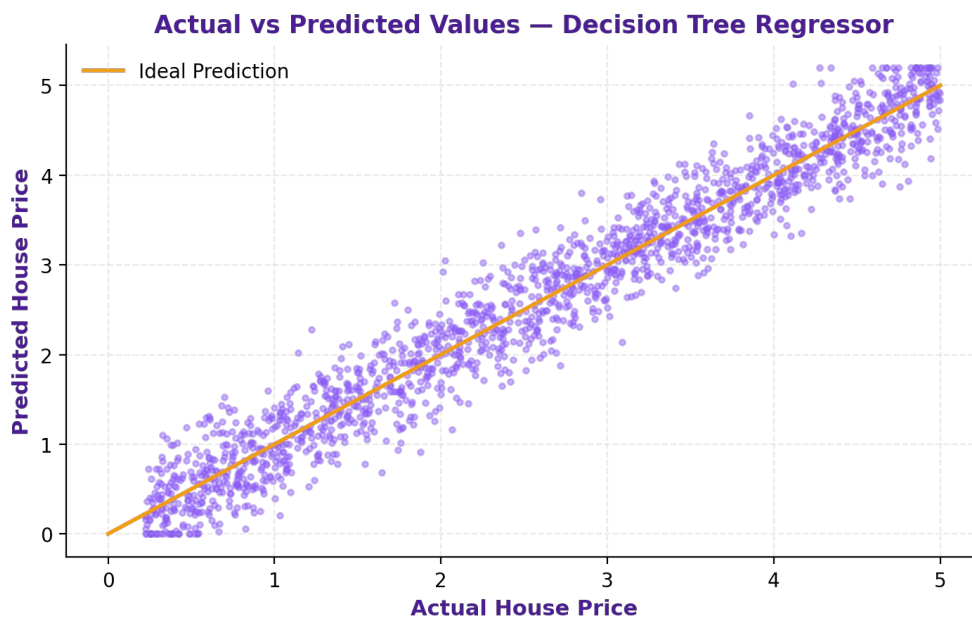


Figure 3 — Actual vs. Predicted house prices for the Decision Tree Regressor, with the ideal-prediction line shown in orange

7 Conclusion

In this project, the House Price Prediction model developed in Task 1 was improved using **feature scaling and model optimization techniques**. Three machine learning models — Linear Regression, Ridge Regression, and Decision Tree Regressor — were implemented and evaluated on the California Housing Dataset.

Based on RMSE and R^2 Score metrics, the **Decision Tree Regressor achieved the best performance** among the tested models. This task strengthened my understanding of feature engineering, model comparison methodology, and the process of selecting the most suitable machine learning model for a given prediction task.

8 References

1. Scikit-learn Documentation — <https://scikit-learn.org>
2. California Housing Dataset Documentation
3. Python Official Documentation — <https://docs.python.org>